Venkata Saratchandra Patnaik Markonda Patnaikuni

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EDUCATION

Master of Science in Computer Science (CGPA: 3.81/4) **December 2025**
Arizona State University, Tempe, AZ.

TECHNICAL SKILLS

Programming & Scripting Languages: Python, R, Java, Kotlin, Go, JavaScript, TypeScript, C
GPU & Parallel Computing: CUDA Programming, PyCUDA, Shared Memory Optimization, Loop Unrolling, Tiling, Grid-Block-Thread Architecture
Web & Mobile Development: React.js, Node.js, Express.js, HTML/CSS, Bootstrap, Android Studio, Flask, REST APIs
Databases: MySQL, PostgreSQL, Google Firebase, MariaDB
Cloud & DevOps Tools: AWS(S3, EKS, EC2, Greengrass, Lambda, Load Balancer), Docker, Kubernetes, Linux, Shell Scripting
CI/CD, Monitoring & Dev Tools: GitHub, GitOps, ArgoCD, Jira, Jenkins, Grafana
Machine Learning & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, OpenAI API

PROFESSIONAL EXPERIENCE

Software Implementation Engineer: Amagi Media Labs, Bengaluru, India **August 2022 - November 2023**

- **Solution Architecture & Automation:** Designed onboarding solutions for content creators, **optimized video ingestion and distribution**, and used **Terraform**-based automation for customer onboarding, reducing manual setup time by **90%.**
- **Cloud & Deployment:** Managed 15+ **containerized Microservices** on **AWS EKS** with **Docker** & **Kubernetes**, increasing deployment frequency by **30%**. Worked extensively in **Linux** environments to deploy, monitor, and troubleshoot systems.
- **Scripting & CI/CD & GitOps:** Deployed **GitOps**-based pipelines via **ArgoCD**, enhancing **CI/CD reliability** and rollback efficiency. Developed **Python** scripts and integrated **APIs** with partner systems for video **preprocessing** and verification before **CDN delivery**, reducing failures by **95%**.

Web Development Intern: Blueplanet Solutions Inc., India **April 2021 - June 2021**

- **Optimized MySQL database** using stored procedures for 'Campus Club', improving query speed by **50%** and enabling faster data access for 1,000+ student profiles.
- **Enhanced a responsive student search UI** with HTML, CSS, and JavaScript, reducing search time by **60%** and improving user experience.
- **Analyzed web server logs** to identify memory leaks, enhancing application stability by 30% and reducing server resource usage.

PROJECTS

Parallel Implementation of Gaussian Filter on GPU (CUDA, PyCUDA, Image Processing)

- Implemented 2D Gaussian convolution on GPU using CUDA & PyCUDA, optimizing performance with thread grids, blocks, and shared memory.
- Applied techniques like loop unrolling, tiling, and separable filters, achieving up to 20× speedup over a CPU-based MATLAB implementation.
- Validated results using PSNR and SSIM metrics; analyzed computation time for various kernel/image sizes to quantify performance gain.

Real-Time Face Recognition on AWS Greengrass (Cloud Computing Project)

- Developed an edge-based face recognition system using AWS IoT Greengrass, MQTT, Lambda, and SQS, simulating smart camera-to-core communication over EC2.
- Deployed MTCNN and FaceNet models under pip-free constraints by integrating raw PyTorch source, optimizing runtime within AWS Greengrass component limits.
- Debugged complex distributed pipeline using CLI tools and logs across Greengrass, MQTT, and Lambda; automated deployments via shell scripts and passed 100+ real-time autograder test cases.